

Summary

This video elucidates core themes, mathematical frameworks, and problem-solving strategies in Probability Theory, focusing on two-event to three-event scenarios, independent events, and the application of addition theorems.

Key takeaways:

- **Event Classification:** "Something is Common" (overlapping) vs. "Nothing is Common" (mutually exclusive)
- **Independent Events:** Complements of independent events are also independent; probabilities multiply
- **Overcounting Prevention:** Use Addition Theorem to subtract intersections
- **Venn Diagram Logic:** Extension from two sets to eight-region three-event systems
- **Problem Applications:** Card drawing, algebraic probability equations, divisibility (1–100), target-hitting scenarios

I. Fundamental Frameworks for Two Events

A. Case Scenarios

1. **Something is Common:** Events overlap ($E1 \cap E2 \neq \emptyset$); use Addition Theorem to adjust for overcounting
2. **Nothing is Common:** Events are disjoint/exclusive ($E1 \cap E2 = \emptyset$)

B. Core Probabilistic Operations

- **Simultaneous Occurrence:** $P(E1 \cap E2)$
- **At Least One Occurs:**
 $P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$
 $P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$
- **Only One Occurs (Exactly One):** Sum of probabilities of (Only E1) and (Only E2)

- **None Occur:**

$$P(E1^c \cap E2^c) = 1 - P(E1 \cup E2) \quad P(E1^c \cap E2^c) = 1 - P(E1 \cup E2)$$

C. Independent Events

- $P(E \cap F) = P(E) \cdot P(F)$
- Complement relationship:
 $P(E^c \cap F^c) = P(E^c) \cdot P(F^c)$
 $P(E^c \cap F^c) = P(E^c) \cdot P(F^c)$
- When solving for unknown probabilities: use $P(E) = x$, $P(F) = y$; do **not** assume $y = 1-x$ unless specified

II. Analysis of Three-Event Systems

A. The Eight-Event Breakdown

Event Type	Logic/Notation	Description
Simultaneous	$E1 \cap E2 \cap E3$	All three events occur
Only One	Only E1, Only E2, Only E3	Exactly one specific event occurs, excluding intersections
Only Two	(E1, E2 only), (E1, E3 only), (E2, E3 only)	Intersections of exactly two events, excluding the third
None	$E1^c \cap E2^c \cap E3^c$	No events occur

B. Addition Theorem for Three Events

$$P(E1 \cup E2 \cup E3) = P(E1) + P(E2) + P(E3) - [P(E1 \cap E2) + P(E2 \cap E3) + P(E1 \cap E3)] + P(E1 \cap E2 \cap E3)$$

$$P(E1 \cup E2 \cup E3) = P(E1) + P(E2) + P(E3) - [P(E1 \cap E2) + P(E2 \cap E3) + P(E1 \cap E3)] + P(E1 \cap E2 \cap E3)$$

III. Practical Problem-Solving Applications

A. Card Probability (Overcounting Example)

- **Problem:** Draw 2 cards from 52. Probability both are Red OR both are Kings

- Red Cards: 26 $\rightarrow P(\text{Both Red}) = 26/52 \times 25/51$
- Kings: 4 $\rightarrow P(\text{Both King}) = 4/52 \times 3/51$
- Overcount Adjustment: Two Kings are also Red $\rightarrow P(\text{Both Red} \cap \text{Both King}) = 2/52 \times 1/51$

B. Algebraic Solution for Independent Events

- Given $P(E \cap F) = 1/12$, $P(\text{Neither } E \text{ nor } F) = 1/2$
1. Equations: $xy = 1/12$, $(1-x)(1-y) = 1/2$
 2. Simplify $\rightarrow x + y = 7/12$
 3. Insight: Use identity $(x-y)^2 = (x+y)^2 - 4xy$ to find linear difference

C. Divisibility and Target Hitting

- **Divisibility (1–100):** Numbers not divisible by 2,3,5 $\rightarrow$ use three-event union
 - $P(\text{Divisible}) = 74/100$
 - $P(\text{Not Divisible}) = 1 - 0.74 = 0.26$ or $13/50$
- **Target Selection (Independent Shooters A,B,C):** Probability (A or B but not C)

$$P(A \cup B) \cdot P(C^c) = [P(A) + P(B) - P(A)P(B)] \cdot [1 - P(C)]$$

$$P(A \cup B) \cdot P(C^c) = [P(A) + P(B) - P(A)P(B)] \cdot [1 - P(C)]$$

IV. Summary of Procedural Insights

1. **Language Sensitivity:** "Only one," "Exactly one," "At least one" dictate formula selection; "Only one" $\rightarrow$ XOR logic
2. **Independence Assumptions:** Probability of "Exactly one" = sum of (first selected, second not) + (second selected, first not)
3. **Complementary Events:** Calculating "None" or "Not" cases is often simpler, especially in divisibility or exclusion problems